# ActivityGraph Gateway Setup

Turning a Raspberry Pi 4B into the network gateway, so every device on the network is measured rather than just the machine doing the capturing.

A laptop on Wi-Fi can only ever see its own traffic. In managed mode a wireless client receives frames addressed to it plus broadcast, and nothing else, so another device's browsing is invisible no matter how long you capture. That is a property of 802.11, not a limitation of the software.

A gateway is different. Every packet from every device must pass through it to reach the internet. Move the capture there and the blind spot disappears, with no change to the code.

### What you are building

```
internet feed
     |
[ Router 1 ]        modem only, Wi-Fi OFF
     |  ethernet
[ Raspberry Pi ]    eth0 = WAN      (built-in port)
     |              eth1 = LAN      (USB adapter)  <-- CAPTURE HERE
     |  ethernet
[ Router 2 ]        access point: DHCP off, static IP
   )  )  )  Wi-Fi
phone / laptops / TV
```

Traffic path: device, router 2, Pi, router 1, internet. Every packet crosses the Pi carrying its original source IP, which is what makes per-device and per-person reporting possible.

> **Read this before you start**
> Three steps in this guide fail *silently*. They produce a dashboard that fills in normally while the data underneath is wrong. They are marked CHECKPOINT. Do not skip the verification at each one.

## Before you start

Beyond the two routers and the Pi, you need:

| Item | Notes |
|---|---|
| microSD card, 16 GB+ | For Raspberry Pi OS |
| Pi power supply | USB-C, 3 A |
| USB Ethernet adapter | Gives the Pi its second network interface |
| USB-A to USB-C adapter | Only if your Ethernet adapter is USB-C. The Pi's USB-C port is power input only and cannot host peripherals. This is the piece most often missing. |
| Two Ethernet cables | Router 1 to Pi, Pi to router 2 |

Time required is roughly 60 to 90 minutes. Your network will be down for a few minutes during Phase 7.

## Phase 1: Flash and boot the Pi

On your laptop, open Raspberry Pi Imager and choose **Raspberry Pi OS Lite (64-bit)**. Before writing, open the settings (gear icon) and configure:

- Hostname: `activitygraph`
- Enable SSH, with a password or your public key
- Your username and password
- Your Wi-Fi details are not needed, since the Pi uses Ethernet

Write the card, put it in the Pi, connect the Pi's **built-in** Ethernet port to a LAN port on router 1, and power it on. Wait about a minute, then from your laptop:

```
ssh <your-user>@activitygraph.local
```

If the hostname does not resolve, find the Pi's address in router 1's DHCP client list and use that instead.

## Phase 2: The USB adapter [CHECKPOINT]

Plug the Ethernet adapter into a **blue USB 3.0 port**, using the USB-A to USB-C adapter if needed. Black ports are USB 2.0 and cap throughput around 280 Mbps. Then:

```
ip -br link
```

You should see `eth0`, `eth1` and `lo`. Most Realtek and ASIX based adapters need no drivers. If `eth1` is missing, check `dmesg | tail` and confirm the port. **Do not continue until it appears.**

### Pin the interface name

Interface names can swap between reboots. Note the adapter's MAC from the output above, then:

```
echo 'SUBSYSTEM=="net", ACTION=="add", ATTR{address}=="AA:BB:CC:DD:EE:FF", NAME="eth1"' \
  | sudo tee /etc/udev/rules.d/10-activitygraph-lan.rules
```

> **Why this matters**
> If eth0 and eth1 swap after a reboot, you capture the WAN side instead of the LAN side. Everything downstream of NAT carries the Pi's own address, so the dashboard would attribute your entire household to a single device. Nothing errors. The numbers just quietly become meaningless.

## Phase 3: Get the code onto the Pi

> **Private repository**
> The Pi cannot clone a private GitHub repo without credentials. Copying from your laptop avoids that entirely and skips the virtual environment, which must be rebuilt for the Pi's architecture anyway.

From your **laptop**:

```
scp -r "C:\Nathan Daily\Study\Projects\ActivityGraph\activitygraph" ^
    <your-user>@activitygraph.local:~/activitygraph
```

> **Check where it actually landed**
> `scp -r src dest` copies *into* dest when that directory already exists, nesting the project one level deeper. On this machine the project root is `~/activitygraph/activitygraph`. Confirm with the find command below; the directory holding `pyproject.toml` is the root, and every later path is relative to it.

Then on the **Pi**:

```
sudo apt update && sudo apt full-upgrade -y
sudo apt install -y python3-pip python3-venv tcpdump dnsmasq nftables

find ~ -name pyproject.toml -not -path '*/.venv/*' 2>/dev/null

cd ~/activitygraph/activitygraph
rm -rf .venv                # the Windows virtualenv is useless on the Pi
python3 -m venv .venv
.venv/bin/pip install -e .

.venv/bin/python -m activitygraph.main --demo
```

The demo run needs no root and no network traffic. If it prints the category and time tables, the software side is working. `tcpdump` is installed because it brings in libpcap and gives you a second tool for checking what an interface is actually receiving.

## Phase 4: Make the Pi a gateway

### 4.1 LAN address

Give `eth1` a static address on its own subnet, kept distinct from router 1's `192.168.1.x` so routing is unambiguous. Leave `eth0` alone: it takes a DHCP lease from router 1 as normal.

> **Check which network manager you have first**
> Raspberry Pi OS switched from dhcpcd to NetworkManager in Bookworm (October 2023). `/etc/dhcpcd.conf` is often still present on such a system: it exists, looks entirely editable, and is completely ignored. Its presence is not evidence that dhcpcd is in use.

```
systemctl is-active NetworkManager && ls -l /etc/dhcpcd.conf
```

**NetworkManager**, which is almost certainly your case on a fresh install:

```
sudo nmcli connection add \
  type ethernet ifname eth1 con-name ag-lan \
  ipv4.method manual ipv4.addresses 192.168.50.1/24 \
  ipv4.never-default yes ipv6.method disabled

sudo nmcli connection up ag-lan
```

- **No gateway is set.** eth1 faces your devices, not the internet. The default route must stay on eth0.
- **ipv4.never-default yes** guarantees eth1 can never become the default route, which protects the capture invariant: an interface carrying the default route is by definition the WAN side.
- **ipv6.method disabled** keeps the LAN single-stack, so devices cannot pick up IPv6 routes that bypass NAT and therefore bypass measurement.

ActivityGraph does capture IPv6 where it is present, so that last setting is belt and braces rather than the only defence. It still matters here: Phase 4.3 sets up forwarding and NAT for IPv4 only, so a device with a working IPv6 route would be routing around the Pi rather than through it, and traffic that never crosses the gateway cannot be measured no matter what the capture layer supports. Expect to see no IPv6 traffic on this build. That is the intended outcome, not a fault.

`sudo nmtui` offers the same thing as an interactive menu if you prefer that to flags.

**dhcpcd**, older images only, appended to `/etc/dhcpcd.conf`:

```
interface eth1
static ip_address=192.168.50.1/24
```

Verify, whichever path you took:

```
ip addr show eth1     # expect inet 192.168.50.1/24
ip route              # default route MUST be on eth0, not eth1
```

On the NetworkManager path, confirm the settings resolved as intended rather than assuming the command that printed "successfully added" did what you meant. `nmcli` accepts unambiguous abbreviations, so a truncated value is accepted silently:

```
nmcli -g ipv4.method,ipv4.addresses,ipv4.never-default,ipv6.method   connection show ag-lan
```

Expected, in order:

```
manual
192.168.50.1/24
yes
disabled
```

Correct any that differ, for example:

```
sudo nmcli connection modify ag-lan ipv6.method disabled
sudo nmcli connection up ag-lan
```

> **NO-CARRIER here is normal**
> At this point `eth1` reports `NO-CARRIER` and its route is marked `linkdown`, because nothing is plugged into it yet. That clears when you cable the access point in Phase 7. The address and route are configured regardless.

### 4.2 DHCP and DNS

`dnsmasq` does two jobs. As a **DHCP server** it hands each device an address and tells it to use the Pi as its gateway, which is what routes traffic through the capture. As a **DNS forwarder** it resolves names on their behalf.

**Create the config.** Copy and paste this entire block, including the final EOF line. It writes the file in one go, so there is no editor to learn.

```
sudo tee /etc/dnsmasq.d/activitygraph.conf > /dev/null <<'EOF'
# Serve only on the LAN side, never on eth0 which faces the router.
interface=eth1
bind-interfaces

# Addresses for your devices. .1 to .49 stay free for fixed addresses
# such as the Pi (.1) and the access point (.2).
dhcp-range=192.168.50.50,192.168.50.200,12h

# Tell clients the Pi is their gateway and DNS server.
dhcp-option=option:router,192.168.50.1
dhcp-option=option:dns-server,192.168.50.1

domain=lan
log-queries
log-facility=/var/log/dnsmasq.log
EOF
```

**Apply it, then check it actually started.** A config error makes dnsmasq exit rather than run with defaults, so this is worth verifying:

```
sudo systemctl restart dnsmasq
systemctl is-active dnsmasq || sudo journalctl -u dnsmasq -n 20 --no-pager
```

**Confirm it is listening on the LAN side only.** Use ss's own port filter rather than grepping the text, which would also match unrelated ports such as 5353 and 53412:

```
sudo ss -lnup 'sport = :53'
```

Expect three lines, all belonging to dnsmasq:

| Address | Meaning |
|---|---|
| 192.168.50.1:53 | The LAN side. This is the one that matters |
| 127.0.0.1:53 | Loopback, so the Pi resolves names for itself. Normal |
| [::1]:53 | The IPv6 loopback equivalent. Also normal |

Two things should **not** appear. `0.0.0.0:53` means bind-interfaces is not taking effect and the Pi is offering DNS to every network it touches. Your eth0 address, the 192.168.1.x one, means it is serving the upstream network, where a second DHCP and DNS server causes real problems.

Anything on port 5353 is `avahi-daemon` doing mDNS. It is unrelated to this setup and can be ignored.

> **If dnsmasq will not start: address already in use**
> Another process holds port 53, usually systemd-resolved. Run `sudo systemctl disable --now systemd-resolved` and then `sudo systemctl restart dnsmasq`.

Two settings are load-bearing. `bind-interfaces` keeps dnsmasq off eth0, so the Pi never offers DHCP upstream where a second DHCP server would cause real problems. The `option:router` line is what makes devices route through the Pi at all: without it they may receive the wrong gateway and bypass the capture entirely.

### 4.3 Forwarding and NAT

Two separate things are needed, and the Pi does neither by default.

- **Forwarding** lets the Pi pass packets between its two interfaces. Linux refuses by default, because a normal computer should not silently relay other machines' traffic.
- **NAT (masquerade)** rewrites outgoing packets to appear to come from the Pi. The upstream router knows nothing about your 192.168.50.x devices, so without this their replies have nowhere to return to.

**Enable forwarding**, permanently across reboots:

```
echo 'net.ipv4.ip_forward=1' | sudo tee /etc/sysctl.d/99-activitygraph.conf
sudo sysctl -p /etc/sysctl.d/99-activitygraph.conf

cat /proc/sys/net/ipv4/ip_forward     # expect 1
```

**Back up any existing firewall rules** before overwriting them:

```
[ -f /etc/nftables.conf ] && sudo cp /etc/nftables.conf /etc/nftables.conf.backup
```

**Write the ruleset.** Paste the whole block including the final EOF:

```
sudo tee /etc/nftables.conf > /dev/null <<'EOF'
#!/usr/sbin/nft -f
flush ruleset

table inet filter {
  # Allow traffic to pass between eth1 (devices) and eth0 (internet).
  chain forward { type filter hook forward priority 0; policy accept; }
}
```

```
table ip nat {
  chain postrouting {
    type nat hook postrouting priority 100; policy accept;
    # Rewrite outgoing packets to come from the Pi's WAN address.
    oifname "eth0" masquerade
  }
}
EOF
```

**Apply it, make it load on boot, and verify it parsed:**

```
sudo systemctl enable --now nftables
systemctl is-active nftables || sudo journalctl -u nftables -n 20 --no-pager
sudo nft list ruleset
```

The last command must show both the `forward` chain and the `masquerade` line. An empty ruleset means the file did not parse, even though `enable --now` reported success.

The printed ruleset will not match your file character for character, and that is expected:

| You wrote | nft displays |
|---|---|
| priority 0 | priority filter |
| priority 100 | priority srcnat |

Those are the symbolic names for those numeric priorities. Your comments are also absent, because nftables does not retain them in the loaded ruleset. Neither indicates a problem.

Nothing routes yet, because no devices are connected. You test this for real at the end of Phase 8.

## Phase 5: Router 2 as an access point

> **Configure this router in isolation**
> Both routers ship on `192.168.1.1` with the same default SSID and admin password. Power router 1 OFF while you do this. With both live you get an address conflict, and a real chance of reconfiguring the wrong box without noticing.

Connect your laptop to router 2 and open `http://192.168.1.1`.

### The easy path

Look for an **Operation Mode** setting offering **Access Point** alongside **Wireless Router**. Selecting Access Point does most of the following in one toggle and is far less error-prone.

### The manual path

- **DHCP server: OFF.** The Pi is the DHCP server now. Two DHCP servers on one segment hand out conflicting leases and cause intermittent failures that are painful to diagnose.
- **LAN IP: static 192.168.50.2**, subnet 255.255.255.0. This must sit outside the 192.168.50.50 to .200 DHCP range configured earlier.
- **SSID: something distinct**, for example Home-AG. Do not reuse the old name.
- **Wi-Fi: ON.** This is the radio your devices will use.
- Its DSL and WAN ports stay unused. You are using only its switch and radio.

**Beware the advice you will find online**
Searching for how to reuse a router as an access point usually returns a double-NAT method: connect its **WAN** port and give it a different subnet. That works for extending Wi-Fi and is wrong here. It NATs every device before the traffic reaches the Pi, so the capture sees one address instead of your actual devices.

### Where these settings live

The DHCP toggle is frequently **not** on a page of its own. On TP-Link gateways, and many others, both settings sit on the same page, with the checkbox below the IP fields:

```
Advanced > Network > LAN Settings

    IP Address      192.168.50.2
    Subnet Mask     255.255.255.0
    DHCP Server     [ ] Enable      <- checkbox, below the IP fields
```

Make sure you are in the **Advanced** view rather than Basic, which usually hides LAN settings entirely. Changing both on one page is convenient: save once and the router drops off the network a single time rather than twice.

**Saving disconnects you, and that is expected**
The admin page is no longer at the address your browser is using. Reach it again at 192.168.50.2 once the Pi is serving DHCP in Phase 8, or immediately by giving your computer a temporary static address such as 192.168.50.10, mask 255.255.255.0, no gateway.

## Phase 6: Router 1 becomes a modem

Power router 1 back on, open its admin page, and **turn off its Wi-Fi**, both the 2.4 GHz and 5 GHz radios. Leave everything else alone: it keeps the DSL connection and continues handing out addresses on `192.168.1.x`, which is how the Pi's `eth0` gets its address.

**The single most common failure**
Router 1 sits *above* the Pi. Any device that associates with its Wi-Fi reaches the internet without the Pi ever seeing it, and shows no traffic at all. If you also reused its SSID on router 2, devices may silently reattach to the wrong one.

## Phase 7: Cabling

**Do not change how router 1 reaches the internet**
It is your existing working router, so whatever is plugged into it now is already correct. Leave that cable exactly as it is. The only change router 1 needs is its Wi-Fi turned off.

| From | To |
|---|---|
| Your internet feed, unchanged | Router 1. A thin phone cable means the **DSL** port (FTTN or FTTB). An Ethernet cable from an NBN box or fibre NTD means the **WAN** port. Never a LAN port. |
| Router 1, any LAN port | Pi, built-in Ethernet (eth0) |

| From | To |
|---|---|
| Pi, USB adapter (eth1) | Router 2, any **LAN** port |

Only the last two rows are new. The connectors are not interchangeable: a DSL port is RJ11, narrower with 6 pins, while WAN and LAN ports are RJ45, wider with 8 pins. An Ethernet plug will not seat properly in a DSL socket.

> **Use router 2's LAN port, never its WAN port**
> Cabling into the WAN port makes router 2 route and NAT the traffic before it reaches the Pi. That destroys per-device visibility in exactly the same way capturing on eth0 would.

## Phase 8: Verify the plumbing [CHECKPOINT]

Connect your phone to router 2's Wi-Fi, then look at its IP address in the Wi-Fi settings.

| Phone shows | Meaning |
|---|---|
| `192.168.50.x`, gateway `192.168.50.1` | Correct. Traffic is routing through the Pi. |
| `192.168.1.x` | Wrong. It is talking to router 1 and bypassing the Pi. Return to Phase 6 and confirm that radio is off. |

On the Pi, confirm the lease was issued:

```
cat /var/lib/misc/dnsmasq.leases
```

Your phone should appear with its hostname. Confirm it also has working internet before continuing.

## Phase 9: Run ActivityGraph

```
cd ~/activitygraph
sudo .venv/bin/python -m activitygraph.main \
  --serve --serve-source live -i eth1 \
  --host 0.0.0.0 --no-browser --dhcp-leases \
  --accounts ~/activitygraph/accounts.db \
  --history-db ~/activitygraph/history.db
```

| Flag | Why |
|---|---|
| `-i eth1` | The LAN side. **Never eth0.** ActivityGraph warns at startup if the interface you pass carries the default route. |
| `--dhcp-leases` | Names and registers every device from the Pi's own lease table. Works with any router, needs no credentials. |
| `--host 0.0.0.0` | Needed to reach the dashboard from another machine. It also publishes it to the whole LAN, which is why the flag below is on the same line. |
| `--accounts` | Requires people to sign in, and shows each of them only the devices they have claimed. Without it the dashboard is unauthenticated and anyone who can reach the port can read every device's history. |

| Flag | Why |
|---|---|
| `--history-db` | Records activity over time, and stores hostnames learned from DNS so the service breakdown survives a reboot. A durable record of what every device did: treat it as private. |

Open `http://192.168.50.1:8420` from any device on the network. Browse something distinctive on your phone, wait roughly 10 seconds for flows to close and classify, then check **Where Your Time Went** and **Time by Person**.

### 9.1 Create the admin account first

> **Whoever registers first becomes the admin**
> The first account created can see every device on the network, and there is no other gate on it. Register yours immediately, before anyone else opens the dashboard.

Everyone else then registers their own account and starts out seeing nothing at all. They add a device by opening the dashboard **on that device** and pressing **Claim this device** in the My Devices panel.

Claiming is checked against the connection's own source address, so a phone can only be added from the phone. That is what makes the dashboard safe to hand out: nobody can attach a device they are not holding, and nobody needs to be told anyone else's MAC address. A device belongs to one person at a time, so shared items such as a TV are usually left unclaimed, which leaves them visible to the admin only.

The command above gives the account database an explicit path so that the service in Phase 10 uses the same one. Left bare, `--accounts` defaults to `~/.activitygraph/`, which resolves to root's home under `sudo`, and you would find yourself registering all over again once the service took over. The session key sits beside the database; deleting it signs everyone out. If you forget the admin password, delete both files and register again: that discards accounts and claims, but not captured history.

## Phase 10: Grouping and permanence

### 10.1 Grouping devices without accounts

With `--accounts` on, people build their own groups by claiming devices and this step is unnecessary. The file-based alternative still exists for a gateway you administer alone, or if you would rather assign devices centrally than ask anyone to sign in.

Take the MAC addresses from the Devices panel and create `~/groups.json`:

```
{
  "Nathan": ["AA:BB:CC:00:00:01", "AA:BB:CC:00:00:02"],
  "Guest":  ["AA:BB:CC:00:00:03"]
}
```

Add `--device-groups ~/groups.json` to the command. Devices you do not list appear under Ungrouped, which is where shared items such as a TV naturally belong. The two mechanisms can run together: groups label the Time by Person panel, accounts decide who is allowed to see what.

### 10.2 Run it on boot

Set a variable to your project root first, so the paths are filled in for you rather than edited by hand:

```
AG=$(dirname "$(find ~ -name pyproject.toml -not -path '*/.venv/*' | head -1)")
echo "$AG"        # sanity check: this is your project root
```

Then paste the whole block, including the final EOF:

```
sudo tee /etc/systemd/system/activitygraph.service > /dev/null <<EOF
[Unit]
Description=ActivityGraph
After=network-online.target
Wants=network-online.target

[Service]
Type=simple
User=root
WorkingDirectory=$AG
ExecStart=$AG/.venv/bin/python -m activitygraph.main --serve --serve-source live -i eth1 --host 0.0.0.0 -
Restart=always
RestartSec=10

[Install]
WantedBy=multi-user.target
EOF
```

Two deliberate choices there. The heredoc is unquoted so $AG is replaced with your real path as the file is written. And ExecStart is one long line rather than using backslash continuations, because a backslash inside an unquoted heredoc is consumed by the shell before systemd ever sees it, producing a mangled command.

**Check the result, then enable it:**

```
cat /etc/systemd/system/activitygraph.service

sudo systemctl enable --now activitygraph
systemctl is-active activitygraph || journalctl -u activitygraph -n 30
```

> **Systemd does not expand the ~ shortcut**
> Every path in the unit must be absolute. Confirm the file you just printed contains no $AG and no ~ characters. Note also that `pi` is no longer the default username on current Raspberry Pi OS images.

## Troubleshooting

| Symptom | Most likely cause |
|---|---|
| Phone gets a 192.168.1.x address | It associated with router 1's Wi-Fi. Turn that radio off. |
| Phone absent from dnsmasq.leases | Router 2's DHCP server is still on, or it is cabled by its WAN port instead of a LAN port. |
| Leases fine, but no traffic recorded | Capturing the wrong interface. It must be eth1. |
| Everything attributed to one IP address | Capturing eth0 post-NAT, or eth0 and eth1 swapped after a reboot. Check the udev rule from Phase 2. |
| Traffic appears but has no categories | Flows classify when they close. Allow about --idle-timeout seconds. |
| Dashboard unreachable | Confirm --host 0.0.0.0 and that you are browsing to 192.168.50.1, not the old 192.168.1.x address. |

| Symptom | Most likely cause |
|---|---|
| No internet anywhere | Check IP forwarding and the nftables masquerade rule from Phase 4.3. |
| Signed in, but every panel is empty | A non-admin account sees only the devices it has claimed. Open the dashboard on the device itself and claim it. |
| A lot of traffic filed as plain HTTPS | Services are identified from DNS, which the Pi only sees once devices start querying it. It fills in over the first few minutes, and --history-db carries it across reboots. |
| Everyone logged out after a reboot | The session key moved. It lives beside the path given to --accounts, so keep that path the same between the manual run and the service. |
| A phone shows as a bare model name | It announced no better name. Devices vary in what they broadcast; add an alias with --device-aliases if you want to override it. |
| No IPv6 traffic at all | Expected. The LAN is single-stack by design. See Phase 4.1. |

## Things to know

- **The Pi becomes a single point of failure.** If it is off, the network is down. Restart=always covers crashes, not power loss or SD card corruption.
- **scapy drops packets under sustained load.** Around 100 Mbps this is unlikely, but a saturated link can outrun Python packet parsing. Time in categories degrades gracefully; precise byte accounting does not.
- **Device-to-device traffic may bypass the Pi.** A phone casting to a TV, where both sit on router 2, is switched locally and never reaches the gateway. Only traffic crossing the Pi is visible.
- **You will lose SSH access briefly** during Phase 7, because your laptop is still on the old network. Move it to router 2's Wi-Fi and reach the Pi at 192.168.50.1.
- **The admin account sees everyone's activity.** Accounts stop household members reading each other's browsing; they hide nothing from whoever holds the admin login, or from anyone with shell access to the Pi. If you are setting this up for other people, tell them that rather than letting them assume otherwise.

### Rollback

Nothing here is permanent. Turn router 1's Wi-Fi back on, reconnect your devices to it, and power the Pi off. Neither router was changed in a way that cannot be undone from its admin page.